

Homework (Carolina Reaper)

- Q1.** A music festival has a parabolic stage with equation $y = x^2 + 4x + 4$. What is the vertex of the parabola?
 (A) $(-2, 0)$
 (B) $(-2, 4)$
 (C) $(0, 4)$
 (D) $(-4, 0)$
 (E) $(-4, 4)$
- Q2.** A painter is creating a mural on a canvas that has a quadratic shape with equation $y = -x^2 + 5x - 3$. What is the axis of symmetry?
 (A) $x = -1$
 (B) $x = 1$
 (C) $x = 2.5$
 (D) $x = 3$
 (E) $x = 5$
- Q3.** A sound wave has a frequency that follows the quadratic equation $y = 2x^2 - 5x + 1$. What is the maximum frequency?
 (A) $y = -1.25$
 (B) $y = 0.5$
 (C) $y = 1$
 (D) $y = 2$
 (E) $y = 3$
- Q4.** A photographer is taking a picture of a parabolic curve with equation $y = x^2 - 2x - 3$. What are the x-intercepts?
 (A) $x = -1, x = 3$
 (B) $x = -2, x = 2$
 (C) $x = 0, x = 4$
 (D) $x = 1, x = 5$
 (E) $x = -3, x = 1$
- Q5.** A rhythm pattern has a quadratic shape with equation $y = -2x^2 + 3x + 1$. What is the direction of the parabola?
 (A) Up
 (B) Down
 (C) Left
 (D) Right
 (E) No direction
- Q6.** A canvas has a quadratic shape with equation $y = x^2 + 2x - 1$. What is the minimum value of the quadratic?
 (A) $y = -2$
 (B) $y = -1$
 (C) $y = 0$
 (D) $y = 1$
 (E) $y = 2$
- Q7.** A sound frequency has a quadratic equation $y = x^2 - 4x + 3$. What is the axis of symmetry?
 (A) $x = -2$
 (B) $x = -1$
 (C) $x = 2$
 (D) $x = 3$
 (E) $x = 4$
- Q8.** A painter is mixing colors on a quadratic curve with equation $y = -x^2 + 2x + 1$. What are the y-intercepts?
 (A) $y = -1$
 (B) $y = 0$
 (C) $y = 1$
 (D) $y = 2$
 (E) $y = 3$

Open Ended Questions (FRQ)

- Q9.** A music composer is creating a quadratic melody with equation $y = x^2 - 3x + 2$. Graph the parabola, identify the vertex, axis of symmetry, and x-intercepts, and analyze the maximum and minimum values.
- Q10.** A canvas artist is designing a quadratic pattern with equation $y = -2x^2 + 5x - 1$. Find the x-intercepts, identify the direction of the parabola, and determine the minimum value of the quadratic.

Chef's Tips

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- Tip 1: Emphasize the importance of graphing and analyzing quadratic equations.
- Tip 2: Encourage students to use real-world applications, such as music and art, to understand quadratic concepts.
- Tip 3: Remind students to check their work and provide clear explanations for their answers.